

RESTAURANT GROWTH POTENTIAL MODELING AND STRATEGIC CLASSIFICATION SYSTEM

Submitted By:

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Business Analytics

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Unified Mentor for providing me with the opportunity to work on this internship project. This project helped me understand practical applications of business analytics, data analysis, and strategic decision-making. I also gained valuable experience working with Python, Google Colab, and data visualization tools. This internship has significantly improved my analytical and technical skills.

ABSTRACT

This project focuses on analysing restaurant business performance using data analytics techniques. The objective is to identify restaurants with different levels of growth potential by analysing operational and financial performance indicators. The dataset includes restaurant revenue, delivery performance, customer ordering behaviour, and profit-related metrics. By performing data cleaning, statistical analysis, revenue evaluation, profit calculation, and strategic classification, restaurants were categorized into High Growth, Medium Growth, and Low Growth groups. The project demonstrates how data-driven insights can support business strategy and decision-making in the restaurant industry.

PROBLEM STATEMENT

The restaurant industry operates in a highly competitive environment where business growth depends on customer demand, operational efficiency, delivery systems, and financial performance. Modern restaurants use multiple sales channels such as in-store dining, self-delivery services, and online delivery platforms. Business owners require analytical systems to evaluate restaurant performance and identify growth opportunities. Data analytics helps businesses understand customer behavior, revenue generation, and profit trends, enabling better strategic decisions for long-term sustainability.

DATASET DESCRIPTION

The dataset used in this project contains restaurant business performance information collected from multiple operational and financial sources. It includes data related to monthly customer orders, in-store revenue, online delivery revenue, delivery platform performance, and net profit generation. The dataset consists of multiple variables that represent both operational efficiency and financial performance. These variables were analysed to understand restaurant behavior and identify factors contributing to business growth.

DATASET FIELDS UTILIZED (HIGH-DIMENSIONAL)

The following important dataset fields were utilized during project analysis:

- **MonthlyOrders** – Total number of customer orders placed monthly
- **InStoreRevenue** – Revenue generated through direct restaurant sales
- **UberEatsRevenue** – Revenue generated through UberEats delivery platform
- **DoorDashRevenue** – Revenue generated through DoorDash platform
- **SelfDeliveryRevenue** – Revenue generated through self-delivery services
- **InStoreNetProfit** – Profit earned through direct in-store business
- **UberEatsNetProfit** – Profit earned from UberEats platform
- **GrowthFactor** – Performance score representing restaurant growth potential
- **TotalRevenue** – Combined revenue calculated during project analysis
- **TotalProfit** – Combined profit calculated during analysis
- **GrowthCategory** – Final restaurant classification output generated by the model

DATA SCIENCE METHODOLOGY (STEP-BY-STEP)

The project followed a structured data analytics methodology for extracting business insights from the restaurant dataset.

Step 1: Data Collection

The restaurant dataset was imported into Google Colab using Python libraries.

Step 2: Data Cleaning

The dataset was inspected for missing values and inconsistencies using null-value analysis.

Step 3: Exploratory Data Analysis

Statistical summaries were generated to understand business performance trends.

Step 4: Revenue Analysis

Revenue from multiple channels including in-store sales and online delivery platforms was combined for analysis.

Step 5: Profit Analysis

Net profit metrics were analyzed to identify financially strong restaurant units.

Step 6: Strategic Classification

Restaurants were categorized into High Growth, Medium Growth, and Low Growth groups based on growth factor values.

Step 7: Data Visualization

Bar charts and histograms were generated for visual business analysis.

Step 8: Final Output Generation

The processed dataset was exported for final reporting and strategic business interpretation.

DATA ANALYSIS AND FINDINGS

After performing data analysis, several important business insights were identified. Revenue analysis showed that restaurants using multiple sales channels generated higher overall revenue compared to restaurants depending only on in-store sales. Profit analysis indicated that financially strong restaurants maintained higher profitability across delivery platforms. Customer order analysis helped identify restaurants with strong customer demand patterns. Visualization techniques showed variations in revenue distribution and monthly order behaviour. These findings helped identify business performance differences between restaurant units.

DATA VISUALIZATION

GROWTH CATEGORY GRAPH

Growth Category Analysis

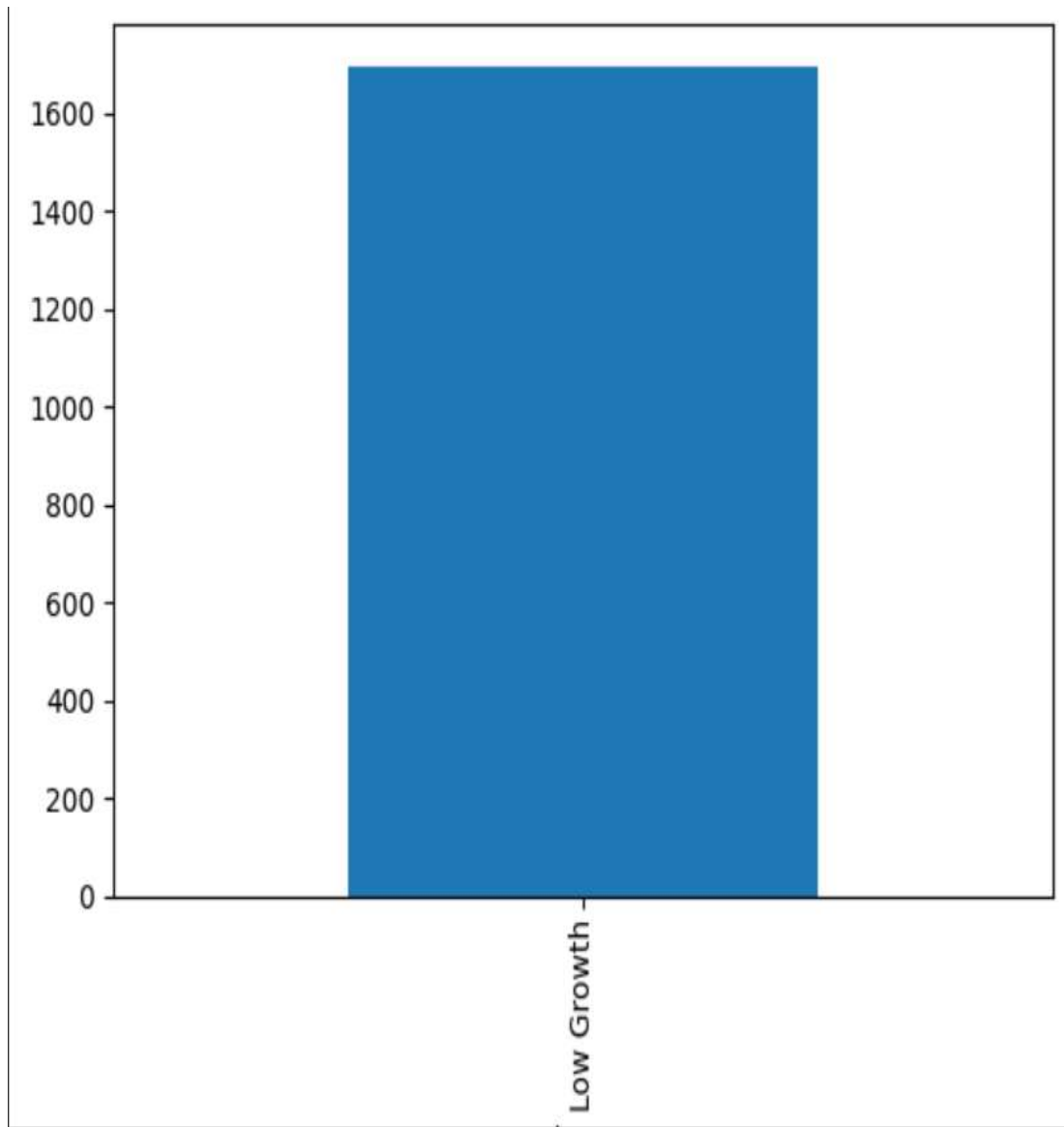


Figure 1: Restaurant Growth Category Classification

REVENUE DISTRIBUTION

Revenue Distribution Analysis

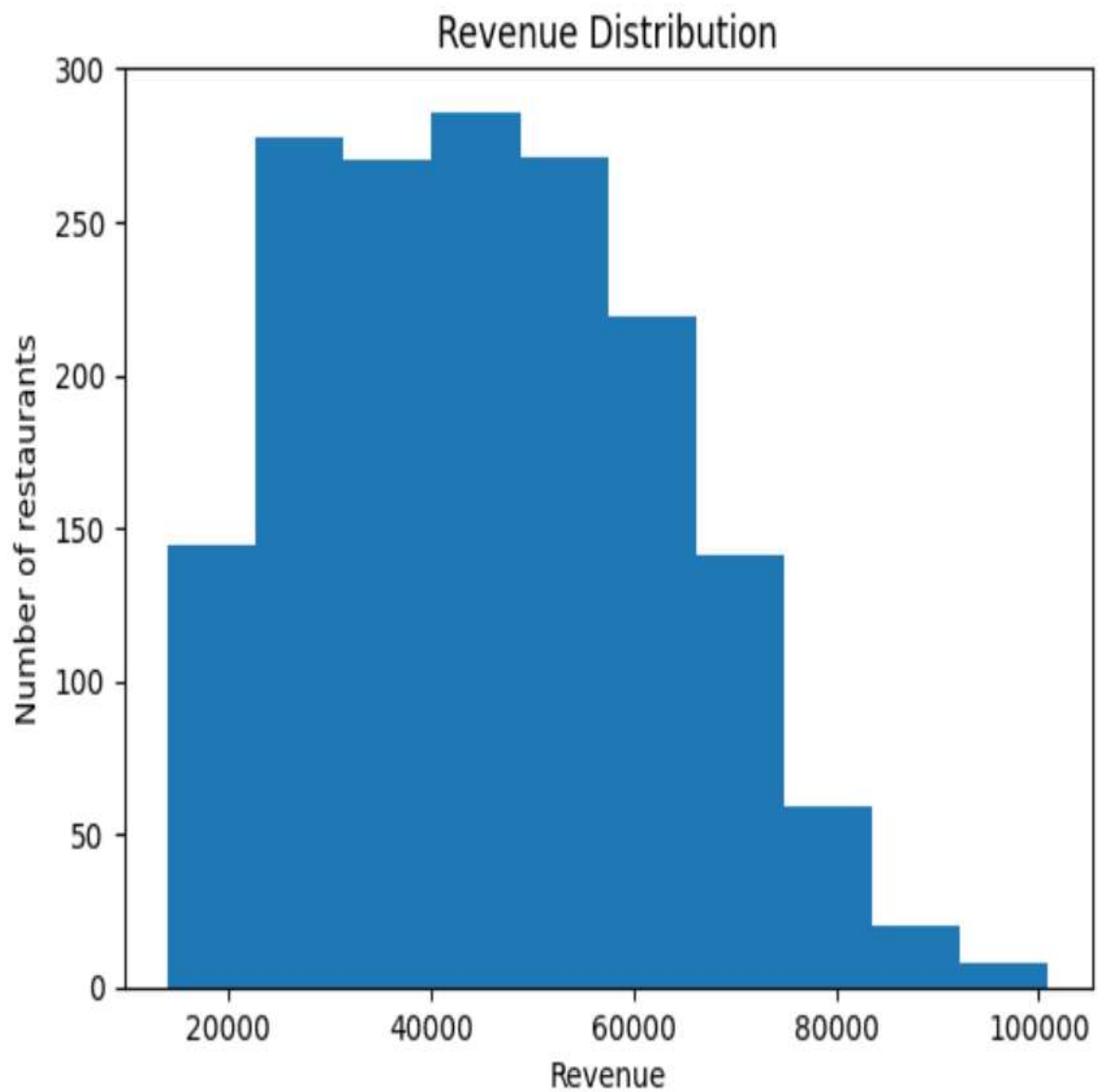


Figure 2: Revenue Distribution Across Restaurants

MONTHLY ORDERS DISTRIBUTION

Monthly Orders Analysis



Figure 3: Monthly Customer Orders Distribution

STRATEGIC CLASSIFICATION OUTPUT

One of the primary objectives of this project was to build a strategic classification system for restaurant growth potential. Based on the GrowthFactor variable, restaurants were categorized into three major business categories.

- High Growth Restaurants – Restaurants with strong business performance and high future growth potential.**
- Medium Growth Restaurants – Restaurants with stable performance but moderate future growth opportunities.**
- Low Growth Restaurants – Restaurants with comparatively weaker business performance and requiring operational improvement strategies.**

This classification system helps business management identify which restaurants are suitable for expansion, additional investment, or strategic restructuring. The model converts raw business data into actionable decision-making insights.

KEY PERFORMANCE INDICATORS (KPIS)

The following Key Performance Indicators (KPIs) were used during analysis to evaluate restaurant performance:

- **Monthly Customer Orders**
- **Total Revenue Generation**
- **Total Profit Generation**
- **In-Store Sales Performance**
- **Delivery Platform Revenue Performance**
- **Customer Demand Patterns**
- **Growth Factor Score**
- **High Growth Restaurant Percentage**
- **Medium Growth Restaurant Percentage**
- **Low Growth Restaurant Percentage**
- **Multi-Channel Revenue Contribution**
- **Business Profitability Trends**

These KPIs were used to measure operational efficiency and business growth performance.

CONCLUSION

This project successfully demonstrated how data analytics can be applied to solve real-world business problems in the restaurant industry. By analyzing operational and financial business data, valuable insights regarding restaurant performance were identified. Revenue and profit analysis helped measure financial strength, while strategic classification helped categorize restaurants according to growth potential. The project proved that data-driven decision-making can significantly improve business planning, operational strategy, and long-term growth management. This analytical approach can help restaurant businesses make smarter and more effective strategic decisions in a competitive market environment.

TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used to complete this project:

- **Google Colab**
- **Python Programming Language**
- **Pandas Library**
- **NumPy Library**
- **Matplotlib Library**
- **CSV Dataset Processing**
- **Data Cleaning Techniques**
- **Statistical Analysis Methods**
- **Business Analytics Concepts**
- **Data Visualization Techniques**

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